



NOAA
FISHERIES
SWFSC



NOAA Fisheries Glider Challenge:

Accelerating the Use of Passive Acoustic-Equipped Gliders to Monitor and Assess Cetacean Populations in the Pacific

Shannon Rankin

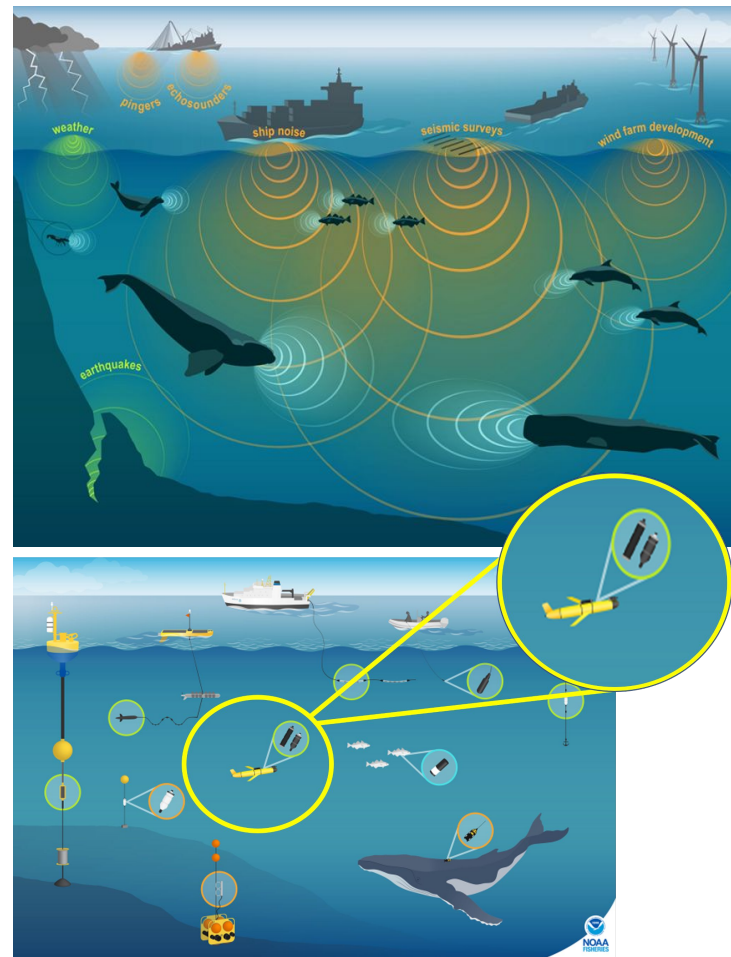
Erin Oleson, Selene Fregosi, Kourtney Burger, Anthony Cossio,
Calla Lloyd-Lim, David K. Mellinger, Erik Norris

NOAA Fisheries UxS-PAM Strategic Initiative

Goal:

PAM-Gliders to augment Traditional Ship Surveys

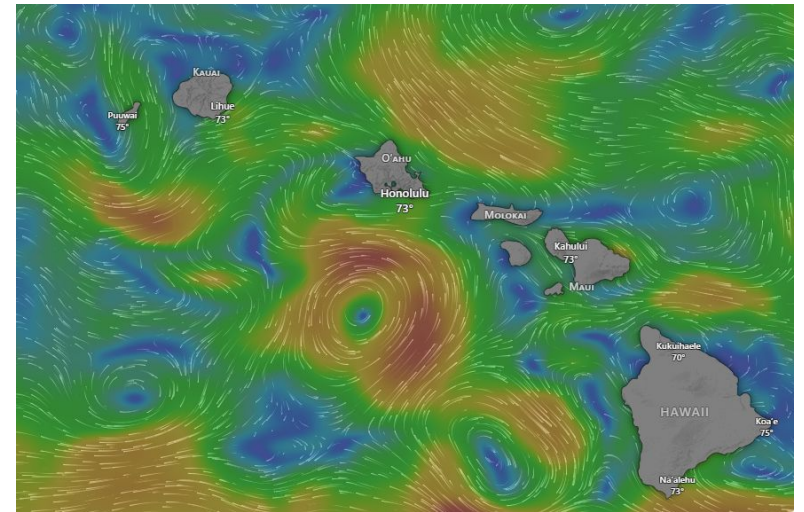
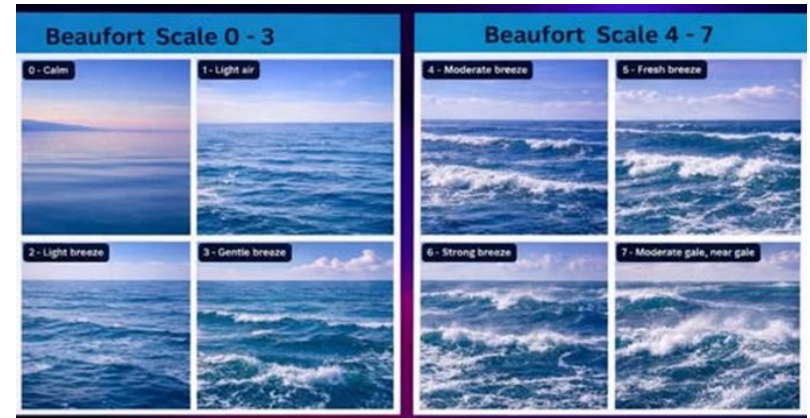
- ☐ Build Capacity to sustain glider operations
- ☐ Concurrent Glider & Shipboard Surveys
- ☐ Develop Spatial Modeling Methods
- ☐ In-Water Instrumentation Testing
(Glider Rodeo!)



Objectives

Field testing of PAM-glider platforms to provide data-driven guidance on the capabilities of each system to:

- Work in different operational environments, and
- Address different research questions.



Approach

1



Trimming

Fine-tune adjustments to prepare for rodeo!

2



FAST Mission Mode

Pilot to prioritize trackline progression

3



SLOW Mission Mode

Pilot to minimize noise while reaching end of line

4



INTERMEDIATE Mission Mode

Balance maintaining course w/ reduced self-noise

5



Drift Mode

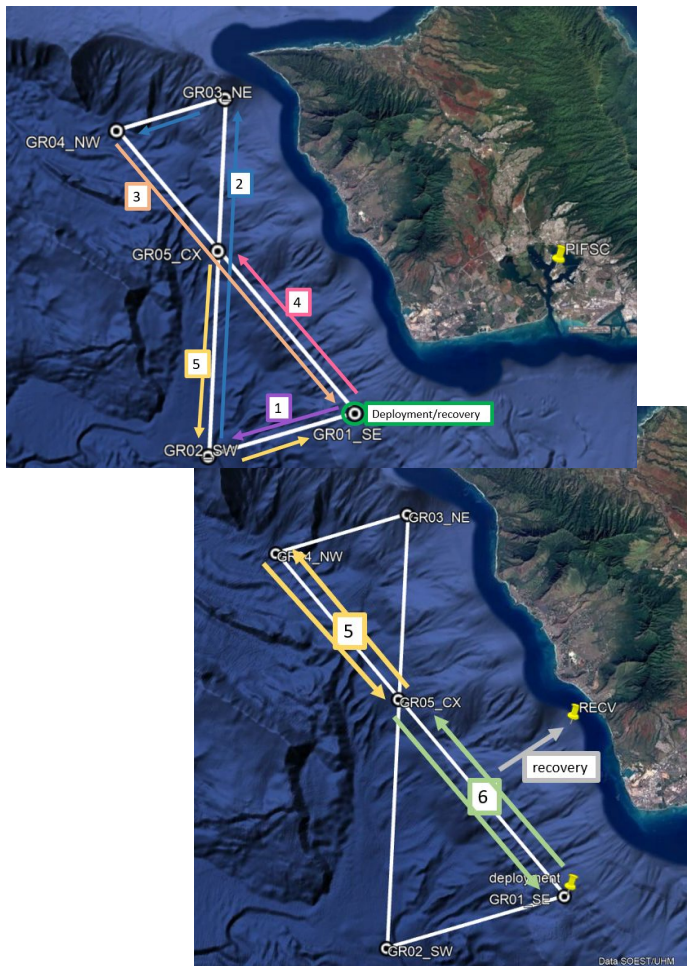
Intermediate mode w/ 20 min drift @ 800m (repeated)

6



Mini Rodeo

Cycle all gliders through Phase 2-5 (12 hrs each)



Assessment Areas

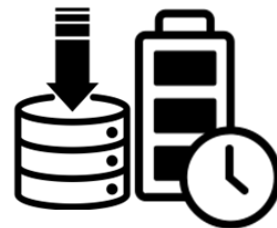


Navigation: Understand navigation strengths & limitations and their impact on endurance, noise, accomplishing mission objectives

Piloting: Understand piloting under different mission modes & their impact on navigation, endurance, noise



Endurance: Assess power and storage endurance under different mission modes



Noise: Assess self-noise and degree to which noise interferes with mission objectives



Rodeo Participants



UW
Seaglider
WISPR



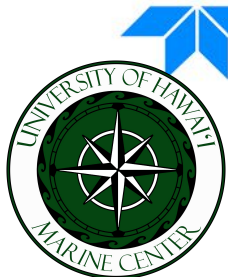
Hefring
Oceanscout
WISPR



Alseamar
SeaExplorer
Auris



Teledyne-Webb
Slocum
Jasco Ocean Observer (1)
DMON (1)
WISPR (1)



TELEDYNE
WEBB RESEARCH
Everywhereyoulook™

HEFRING
ENGINEERING



JASCO
APPLIED SCIENCES

ALSEAMAR
ALCEN



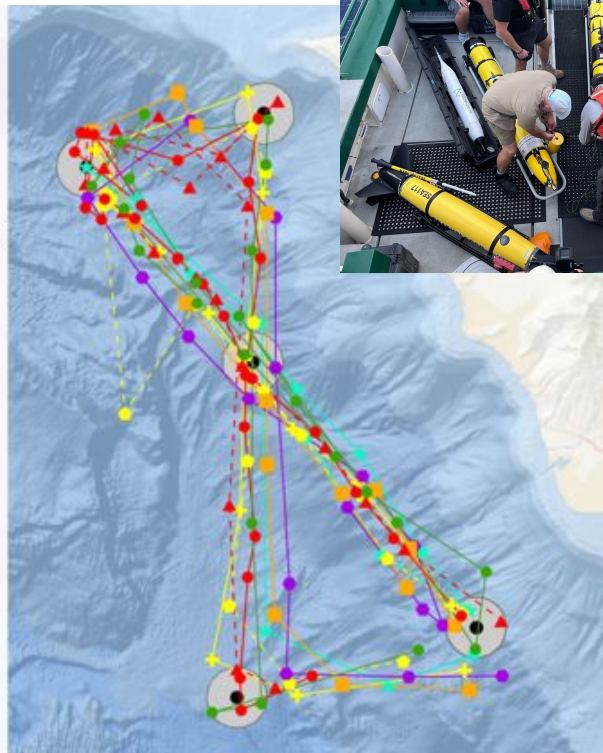
Oregon State
University



Fieldwork

Map of Glider Tracks as of 2/10/2026

- Sea Explorer (Alseamar)
- Ocean Scout (Cornell, Belladonna)
- ▲ Ocean Scout (Cornell, Rosie)
- SeaGlider (OSU)
- SeaGlider (PI)
- + Slocum (SW Risso)
- Slocum (SW Stenella)
- ★ Slocum (Teledyne, Jasco)



Plan: 10 Gliders

Deploy: 8 Gliders

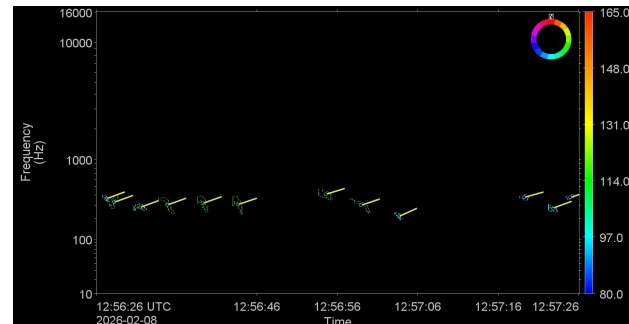
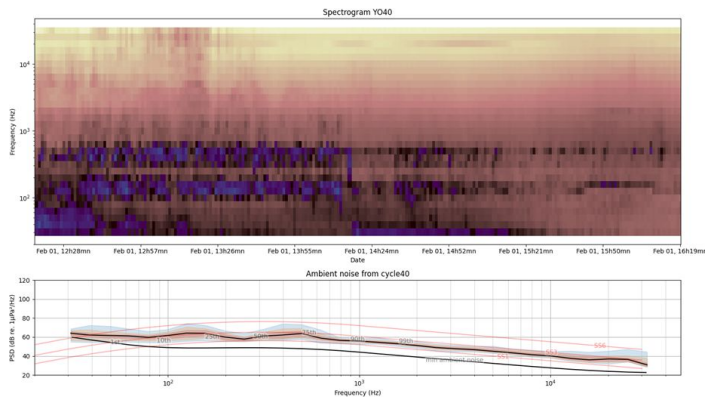
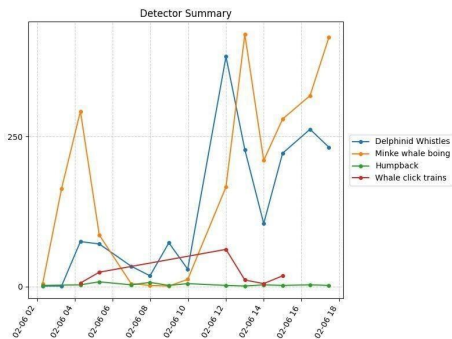
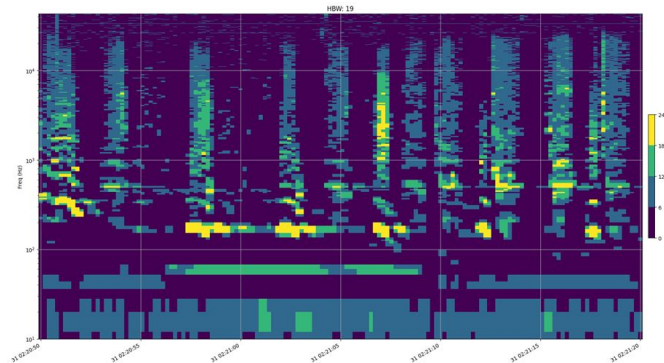
Retrieve: 7 Gliders

Data: 6.5 Gliders!



Data!

- 4.5 Tb Audio Data
- 415 km (per glider)



Analysis Questions



How can we pilot gliders to achieve different mission modes?



How is navigation affected by environmental conditions?



How does Mission Mode affect battery endurance/storage capacity?



How does glider noise interfere with study objectives?



Glider Rodeo Hackweek

SEPTEMBER 15-18 2026

REMOTE ~ ONLINE ~ EVENT !!

NO EXPERIENCE REQUIRED !!



Miss out on the Wet Rodeo?
Join us for the Data Rodeo!



Ideas? Interested? Fill out our survey!

so many thanks to so many people!



UxS Operations Center

Office of Marine and Aviation Operations



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Hackathon Interest Survey



REFOCUS (SWFSC):

REIMAGINING ECOSYSTEM AND FISHERIES OBSERVATIONS BY COMBINING UxS FLEETS

UxSOC AND REFOCUS PROVIDED SWFSC GLIDERS FOR PAM INTEGRATION

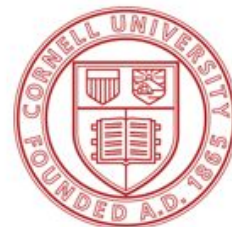
Glider Rodeo Website



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